```
close all
clear
clc
```

# Problem 1

```
syms s11 s22 s33 s23 s13 s12 C S

sigma = [s11 s12 s13;
         s12 s22 s23;
         s13 s23 s33];

Rx = [
    1 0 0;
    0 C S;
    0 -S C
    ];

sigma_x = Rx * sigma * Rx.';
disp("sigma_x:")
disp(simplify(sigma_x))

Ry = [
    C 0 -S;
    0 1 0;
    S 0 C
    ];
sigma_y = Ry * sigma * Ry.';
disp("sigma_y:")
disp(simplify(sigma_y))
```

```
sigma_x:
[          s11,                          C*s12 +
S*s13,                         C*s13 - S*s12]
[C*s12 + S*s13,         s22*C^2 + 2*s23*C*S + s33*S^2, C*(C*s23 + S*s33) -
S*(C*s22 + S*s23)]
[C*s13 - S*s12, C*(C*s23 - S*s22) + S*(C*s33 - S*s23),         s33*C^2 -
2*s23*C*S + s22*S^2]

sigma_y:
[        s11*C^2 - 2*s13*C*S + s33*S^2, C*s12 - S*s23, C*(C*s13 - S*s33) +
S*(C*s11 - S*s13)]
[                         C*s12 - S*s23,
s22,                             C*s23 + S*s12]
[C*(C*s13 + S*s11) - S*(C*s33 + S*s13), C*s23 + S*s12,         s33*C^2 +
2*s13*C*S + s11*S^2]
```

# Problem 2

```
theta1 = deg2rad(90);
theta2 = deg2rad(30);
```

```
theta3 = deg2rad(45);

R1 = [
    1 0 0;
    0 cos(theta1) sin(theta1);
    0 -sin(theta1) cos(theta1)
    ];

R2 = [
    cos(theta2) 0 -sin(theta2);
    0 1 0;
    sin(theta2) 0 cos(theta2)
    ];

R3 = [
    cos(theta3) sin(theta3) 0;
    -sin(theta3) cos(theta3) 0;
    0 0 1
    ];

% Normal matrices
sigma = [
    1 2 5;
    2 3 5;
    5 5 6];

eps = [1 2 5; 2 3 5; 5 5 6];

sigma_prime_R = R3*R2*R1*sigma*R1'*R2'*R3';

% Transformation

c1 = cos(theta1);
c2 = cos(theta2);
c3 = cos(theta3);
s1 = sin(theta1);
s2 = sin(theta2);
s3 = sin(theta3);

Tx = [
    1 0 0 0 0 0;
    0 c1*c1 s1*s1 2*c1*s1 0 0;
    0 s1*s1 c1*c1 -2*c1*s1 0 0;
    0 c1*s1 -c1*s1 c1*c1-s1*s1 0 0;
    0 0 0 0 c1 -s1;
    0 0 0 0 s1 c1
    ];

Ty = [
    c2*c2 0 s2*s2 0 -2*c2*s2 0;
    0 1 0 0 0 0;
    s2*s2 0 c2*c2 0 2*c2*s2 0;
    0 0 0 c2 0 s2;
    c2*s2 0 -c2*s2 0 c2*c2-s2*s2 0;
```

```
    0 0 0 -s2 0 c2
    ];

Tz = [
    c3*c3 s3*s3 0 0 0 2*c3*s3;
    s3*s3 c3*c3 0 0 0 -2*c3*s3;
    0 0 1 0 0 0;
    0 0 0 c3 -s3 0;
    0 0 0 s3 c3 0;
    -c3*s3 c3*s3 0 0 0 c3*c3-s3*s3
    ];

T = Tz*Ty*Tx;

disp("T:")
disp(T)

sigma_vec = [1 3 6 5 5 2]';

sigma_prime = T*sigma_vec;

disp("sigma_prime (T):")
disp(sigma_prime)

disp("sigma_prime (Rotation matrices):")
disp(sigma_prime_R)

% Epsilon


eps_vec = [1 3 6 5 5 2]';
R = diag([1,1,1,2,2,2]);
e_vec = R*eps_vec;

T_eps = R * T / R;


e_prime = T_eps*e_vec;
e_vec_check = (T') \ e_vec;

disp("e_prime (T):")
disp(e_prime)

disp("e_vec_check (T^{-T}):")
disp(e_vec_check)
```

*T:*

| | | | | | |
|---|---|---|---|---|---|
| *0.3750* | *0.1250* | *0.5000* | *0.5000* | *0.8660* | *0.4330* |
| *0.3750* | *0.1250* | *0.5000* | *-0.5000* | *-0.8660* | *0.4330* |
| *0.2500* | *0.7500* | *0.0000* | *-0.0000* | *0.0000* | *-0.8660* |
| *-0.3062* | *0.3062* | *-0.0000* | *-0.6124* | *0.3536* | *0.3536* |
| *0.3062* | *-0.3062* | *-0.0000* | *-0.6124* | *0.3536* | *-0.3536* |
| *-0.3750* | *-0.1250* | *0.5000* | *0.0000* | *0.0000* | *-0.4330* |

```
sigma_prime (T):
   11.4462
   -2.2141
    0.7679
    0.0254
   -2.6136
    1.3840

sigma_prime (Rotation matrices):
   11.4462    1.3840   -2.6136
    1.3840   -2.2141    0.0254
   -2.6136    0.0254    0.7679

e_prime (T):
   11.4462
   -2.2141
    0.7679
    0.0508
   -5.2271
    2.7679

e_vec_check (T^{-T}):
   11.4462
   -2.2141
    0.7679
    0.0508
   -5.2271
    2.7679
```

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